Unit 3

Lesson 4

Beyond Imagination.

Sound Explorers.

Time Frame: [90 min]

Lesson Objectives:

By the end of this lesson students will be able to:

- Identify that sound is made by vibrations.
- Describe how we can feel sound through movement.
- Observe sound moving through classroom objects.
- Use voice commands to turn on/off an LED.

Challenge questions of the lesson

1. What makes sound, and how do we know it's moving?
2. Can we feel sound, even if we can't see it?
3. How does sound travel through things in our classroom?
4. Can we use our voice to control things like lights?

SEL Relation

- **Self-Awareness:** Express excitement and pride in learning how sound helps others.
- **Self-Management:** Practice focus and patience during experiments and activities.
- **Social Awareness:** Show empathy by understanding how technology supports people's needs.
- **Relationship Skills:** Communicate, listen, and collaborate with classmates.
- **Responsible Decision-Making:** Reflect on using sound and technology responsibly to help others.

Engage

Teacher's Role

- **Story Facilitator:** Read the scene aloud or play an audio version. Use tone, gestures, and pauses to bring the story of Lila and Uncle Sam to life. Emphasize emotional shifts (e.g., Uncle Sam's excitement or Lila's pride) to help students connect.
- **Discussion Leader:** Pause after key moments to ask comprehension and reflection questions. Encourage students to predict what will happen next or make connections to real-life uses of voice technology.
- **Think Question Guide:** Lead students through the "Let's Think!" question. Read each choice and let them vote or discuss in pairs before sharing their reasoning. Prompt deeper thinking by asking "Why do you think that's the right answer?"

Expected From Students

- **Active Listening:** Pay attention to the story and respond to questions about how Uncle Sam feels and what the voice commands do.
- **Comprehension & Reflection:** Complete the "Story Reflection" activity by choosing correct answers that show understanding of the narrative.
- **Critical Thinking:** Analyze how voice and sound are used to control smart objects using the "Let's Think!" question. They should explain their answers in their own words.

Parts of the books included


Narrator: The big day had come! Uncle Sam was ready to try out his new smart home. Laila and her family worked hard to set up all the voice controls she had planned.

Uncle Sam: "Wow, Laila! This is amazing! Lights on!"

(The lights turned on, and Uncle Sam smiled happily.)

Laila: "Now you can use your voice to control everything, Uncle Sam!"

Uncle Sam: "This is so cool! I can do things by myself again."

(He moved around the room. He used his voice to open the curtains, turn on the radio, and change the lights.)

Narrator: Uncle Sam felt happy and proud. The smart home-made things easier and helped him feel closer to his family.

Uncle Sam: "Thank you, Laila! My home is better now, and I can do more on my own."

(Laila smiled, feeling proud that she helped her uncle.)

Story Reflection

1. How did Uncle Sam feel when he first used the smart home?
   A. Happy and excited.
   B. Sad and tired.
   C. Angry and confused.

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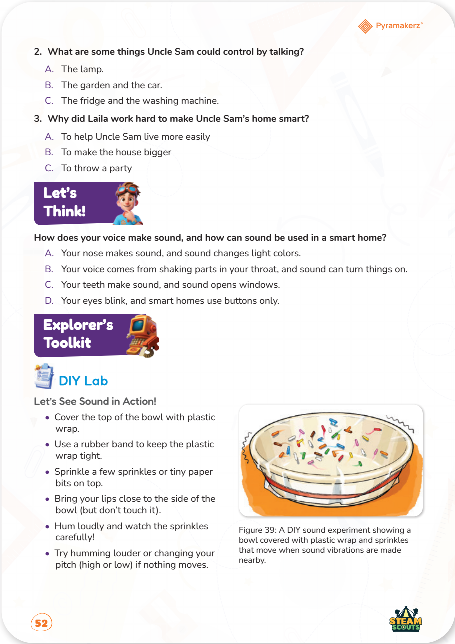
Pyramakerz

2. What are some things Uncle Sam could control by talking?
   A. The lamp.
   B. The garden and the car.
   C. The fridge and the washing machine.
3. Why did Laila work hard to make Uncle Sam's home smart?
   A. To help Uncle Sam live more easily
   B. To make the house bigger
   C. To throw a party

Let's Think!

How does your voice make sound, and how can sound be used in a smart home?
A. Your nose makes sound, and sound changes light colors.
B. Your voice comes from shaking parts in your throat, and sound can turn things on.
C. Your teeth make sound, and sound opens windows.
D. Your eyes blink, and smart homes use buttons only.

Explorer's Toolkit

DIY Lab

Let's See Sound in Action!

- Cover the top of the bowl with plastic wrap.
- Use a rubber band to keep the plastic wrap tight.
- Sprinkle a few sprinkles or tiny paper bits on top.
- Bring your lips close to the side of the bowl (but don't touch it).
- Hum loudly and watch the sprinkles carefully!
- Try humming louder or changing your pitch (high or low) if nothing moves.

Figure 39: A DIY sound experiment showing a bowl covered with plastic wrap and sprinkles that move when sound vibrations are made nearby.

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Story

1. **Set the Context:**
   - Begin by reminding students of Lila and Uncle Sam from previous lessons. Say something like:
     - o "Remember how Lila was trying to help Uncle Sam with a smart home? Today is the big day—let's find out what happens when he tries it for the first time!"
2. **Create Curiosity:**
   - Ask a question to activate thinking:
     - o "What do you think it would feel like to control your home using just your voice?"
   - Invite students to share quick guesses or experiences with voice assistants (e.g., Alexa, Siri).
3. **Read the Story Expressively:**
   - Read the scene with expression and dramatic pauses. Use gestures to act out moments (e.g., turning on lights, speaking commands).

Possible Strategies for Implementation:

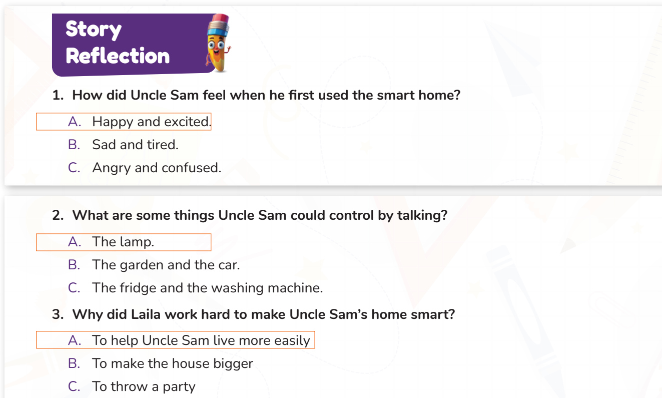
1. **Role-Play Reading:**
   - Assign students to play each character (Narrator, Lila, Uncle Sam) and perform the scene in front of the class. This boosts comprehension, fluency, and engagement.
2. **Story Mapping:**
   - After reading, work together to fill out a simple "Feelings Map":
     - o What problem did Uncle Sam have?
     - o What solution did Lila create?
     - o How did Uncle Sam feel before and after?
3. **Prediction Pause:**
   - Stop midway (before Uncle Sam uses voice commands) and ask students to predict what will happen next.
   - Example prompt: "What do you think he'll say? What might the house do?"
4. **Think-Pair-Share:**
   - After reading, ask: "How did the smart home help Uncle Sam feel more independent?" Let students discuss with a partner, then share as a class.
5. **Anchor to Real Life:**
   - Ask students if they've seen or used smart devices before. Invite them to connect the story to real-world examples.

Story Reflection

Objective:

Students will reflect on the story of Uncle Sam's smart home and answer questions that demonstrate their understanding of how voice-controlled technology can support people's independence and well-being. This activity helps students connect emotions, actions, and technology use, fostering empathy, comprehension, and real-world application of smart solutions that promote accessibility at home.

Answers:



Story Reflection

1. How did Uncle Sam feel when he first used the smart home?
   A. Happy and excited.
   B. Sad and tired.
   C. Angry and confused.
2. What are some things Uncle Sam could control by talking?
   A. The lamp.
   B. The garden and the car.
   C. The fridge and the washing machine.
3. Why did Laila work hard to make Uncle Sam's home smart?
   A. To help Uncle Sam live more easily
   B. To make the house bigger
   C. To throw a party

Possible Strategies for Implementation:

1. Guided Reading Review
   - **How to implement:** Re-read or revisit parts of the story together. Pause at key points where answers are revealed. Ask students to raise their hands or signal when they hear the answer.
2. Think-Pair-Share
   - **How to implement:** For each question, let students first think quietly, then discuss their answer with a partner before sharing with the class. This encourages participation and clarifies understanding.
3. Use Visuals and Gestures
   - **How to implement:** Reenact scenes with student volunteers or use gestures to recall moments (e.g., turning on the light, smiling with pride). Then connect them to the reflection questions.
4. Circle the Correct Answer
   - **How to implement:** Provide students with printed copies and ask them to circle their answers independently. Review together with explanations for each answer.
5. Character Voice Role-Play
   - **How to implement:** Ask students to pretend to be Lila or Uncle Sam and answer a question "in character." For example, "Uncle Sam, how did you feel when the smart home worked?" This deepens empathy and understanding.

Let's Think!

Objective:

Students will explain how sound is produced by vibrations and how voice can be used as an input to operate devices in a smart home. This activity helps students link basic science concepts of sound with real-world applications in accessible technology.

Answers:

Correct Answer:

B. Your voice comes from shaking parts in your throat, and sound can turn things on.

A. Incorrect – the nose is not the main organ responsible for sound production.

C. Incorrect – teeth do not produce sound.

D. Incorrect – smart homes often use voice, not blinking eyes, for control.

Possible Strategies for Implementation:

1. **Think-Pair-Share:**
   - **How to Implement:** After showing the question, ask students to first think silently for 30 seconds. Then, pair up and share their ideas with a partner before answering. Follow with a class discussion to review answers and explain the science behind voice commands and vibrations.
2. **Use a Hands-on Demonstration:**
   - **How to Implement:** Let students gently place their hands on their throats and hum to feel the vibration. Link this tactile experience to the correct answer and explain how microphones in smart homes detect voice vibrations.
3. **Connect to Real Life:**
   - **How to Implement:** Ask students if they've ever seen someone use "Hey Google" or "Alexa." Let them share what they think is happening when a device responds to voice. Use their answers to bridge into the correct choice.
4. **Science Wall Chart Extension:**
   - **How to Implement:** Add a visual poster or chart in the class showing the throat, vocal cords, and how sound travels as vibrations. Revisit the chart during or after answering the question to reinforce understanding.

Explorer's Toolkit

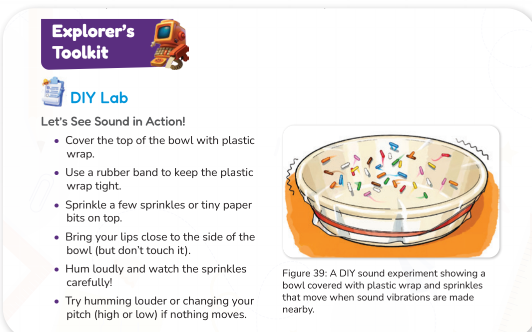
Teacher's Role:

- **Facilitator of Inquiry:** Set up stations and ensure all materials are ready. Prompt students to observe carefully and guide them without giving away the results.
- **Support Observer:** Encourage students to test different sounds and pitches, and ask probing questions like "What do you think caused that?" or "What changed when you hummed louder?"
- **Clarifier:** Help students understand the concept of vibrations by relating the movement of sprinkles to how sound travels through air.

Expected from Students:

- Follow instructions to set up the experiment.
- Make careful observations about the movement of the sprinkles.
- Experiment with different pitches and volumes.
- Record their observations and reasoning for what caused the movement.
- Discuss the results with peers and connect to the idea of sound vibrations.

Parts of the books included



Explorer's Toolkit

DIY Lab

Let's See Sound in Action!

- Cover the top of the bowl with plastic wrap.
- Use a rubber band to keep the plastic wrap tight.
- Sprinkle a few sprinkles or tiny paper bits on top.
- Bring your lips close to the side of the bowl (but don't touch it).
- Hum loudly and watch the sprinkles carefully!
- Try humming louder or changing your pitch (high or low) if nothing moves.

Figure 39: A DIY sound experiment showing a bowl covered with plastic wrap and sprinkles that move when sound vibrations are made nearby.

DIY Lab

Objective:

Students will investigate how sound is caused by vibrations by observing how their humming creates movement in nearby objects. This hands-on experiment allows learners to explore invisible sound waves and connect them to real physical effects.

Materials Needed:

- A bowl
- Plastic wrap
- Rubber band
- Sprinkles or small paper bits
- Quiet classroom space for observation

Activity Instructions:

1. Cover the top of the bowl tightly with plastic wrap and secure it with a rubber band.
2. Sprinkle a few tiny objects (e.g., sprinkles or paper bits) on the plastic wrap.
3. Bring your mouth close to the bowl (without touching it).
4. Hum loudly and observe the movement of the sprinkles.
5. Try humming louder or changing your pitch if nothing moves.
6. Reflect by answering the follow-up questions:

1. What did you see happen when you changed your voice?

The sprinkles jumped a lot

2. What made the sprinkles move?

The sound from my humming

Possible Strategies for Implementation:

1. **Pair-Based Setup: Students work in pairs—one hums while the other watches closely and notes the movement.**
2. **Scientific Journaling: Have students draw the setup in their notebook and record their predictions, results, and reflections.**
3. **Use Visual Aids: Show an animation or video of sound waves before or after the activity to deepen understanding.**
4. **Mini Discussion Circles: Pause mid-activity to let students share early observations and hypotheses.**
5. **Connect to Real Life: Ask students where they've seen or felt vibrations before (e.g., when sitting near a speaker or touching a running machine).**

Explain

Teacher's Role:

- Introduce the concept that sound is made by vibrations, and prepare students to understand how this connects to real-world technology (like voice-controlled devices).
- Facilitate video viewing by setting a clear purpose, guiding key observations, and prompting reflection.
- Support vocabulary development (vibrations, sound waves, control, voice command).
- Clarify misconceptions and encourage students to share their ideas and questions.

Expected From Students:

- Watch the video actively and listen for how sound travels and how it can be used to control devices.
- Participate in pre- and post-video discussions.
- Use simple vocabulary to explain what they learned (e.g., "Sound moves," "It shakes things," "Voice can control lights").
- Begin connecting the science of sound to smart home applications.

Parts of the books included.

- ☐ The bowl

Screen Lessons

Click on the link to watch the video and learn about sound!

https://www.youtube.com/watch?v=m2mxVWrVP58

Screen Lessons

Objective:

Students will explore how sound is made by vibrations and how these vibrations can be used in real-world applications such as smart homes. They will learn to observe and describe how voice commands use sound to trigger responses in electronic systems.

Video Link:

https://www.youtube.com/watch?v=m2mxVWrVP58

Possible Strategies for Implementation:

1. **Pre-Watching: Set a Purpose**
   - **Strategy:** Activate prior knowledge and create curiosity.
   - **How to Implement:**
     - Ask students, “Have you ever turned something on using just your voice?”
     - Write or draw student responses.
     - Tell students to watch the video to find out how sound helps smart homes work.
2. **During Watching: Active Engagement**
   - **Strategy:** Prompt guided observation.
   - **How to Implement:**
     - Pause the video at key moments to ask:
       1. “What is making the object move?”
       2. “Where is the sound coming from?”
     - Use gestures (like tapping the throat or miming vibrations) to help them visualize.

3. **Post-Watching: Discussion and Reflection**

- Strategy: Reinforce understanding through student-led explanation.
- How to Implement:
  - Have students draw what they saw or act out how sound travels and makes things move.
  - Ask, “What did you learn about sound and smart homes?”
  - Encourage pair-share or small group sharing to build language and confidence.

Elaborate and Evaluate

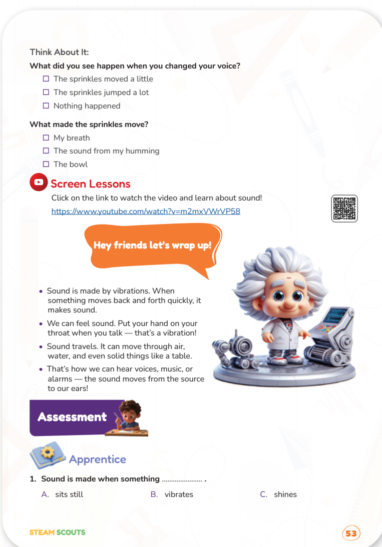
Teacher’s Role:

- Facilitator: Guide students in constructing the string phone step-by-step and explain how vibrations carry sound.
- Safety Monitor: Ensure materials are used safely, especially sharp objects like scissors (if used to poke holes).
- Question Leader: Ask guiding questions like, “Can you hear your friend better when the string is tight?” or “What happens when the string is loose?”
- Connector: Help students connect the activity to real-world concepts, like telephones or how animals communicate.

Expected from Students:

- Work in pairs or small groups to construct a string phone using cups and string.
- Test the string phone and make observations about how well sound travels.
- Identify how vibrations move through the string.
- Reflect on the science behind what they experienced.

Parts of the books included.


Think About It:

What did you see happen when you changed your voice?

- ☐ The sprinkles moved a little
- ☐ The sprinkles jumped a lot
- ☐ Nothing happened

What made the sprinkles move?

- ☐ My breath
- ☐ The sound from my humming
- ☐ The bowl

Screen Lessons

Click on the link to watch the video and learn about sound!

https://www.youtube.com/watch?v=m2mxVWrVP58

- Sound is made by vibrations. When something moves back and forth quickly, it makes sound.
- We can feel sound. Put your hand on your throat when you talk — that’s a vibration!
- Sound travels. It can move through air, water, and even solid things like a table.
- That’s how we can hear voices, music, or alarms — the sound moves from the source to our ears!

Assessment

Apprentice

1. **Sound is made when something ....**

A. sits still B. vibrates C. shines

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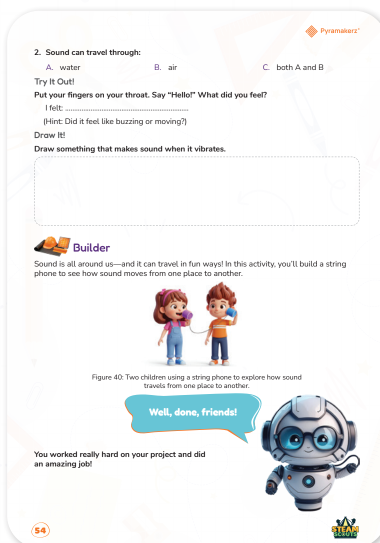
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2. **Sound can travel through:**

A. water B. air C. both A and B

Try It Out!

Put your fingers on your throat. Say “Hello!” What did you feel?

I felt: ....

(Hint: Did it feel like buzzing or moving?)

Draw It!

Draw something that makes sound when it vibrates.

Builder

Sound is all around us—and it can travel in fun ways! In this activity, you’ll build a string phone to see how sound moves from one place to another.

Figure 40: Two children using a string phone to explore how sound travels from one place to another.

You worked really hard on your project and did an amazing job!

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Apprentice

Objective:

Students will demonstrate their understanding that sound is produced by vibrations and can travel through different materials (like air and water). They will also observe and describe physical sensations that accompany sound production using their own voices.

Answers:

1. Sound is made when something ....

B. vibrates

1. Sound can travel through:

C. both A and B

Try It Out – I felt:

- A buzzing or vibrating feeling in their throat (free response; expected answer: "buzzing," "moving," "vibrating," etc.)

Draw It!

Draw something that makes sound when it vibrates.

Possible Strategies for Implementation:

- **Model the Activity First:**
  - o Begin by demonstrating the “Try It Out” section with the whole class. Say "Hello!" loudly and describe what you feel. Let students mirror you.
- **Think-Pair-Share:**
  - o Have students think about what vibration means, pair up to test the activity, and share their answers.
- **Scaffold Vocabulary:**
  - o Pre-teach or review key words like “vibration,” “travel,” and “buzzing” using visuals or real-life examples like guitar strings or loudspeakers.
- **Use Real-Life Examples:**
  - o Connect the concept to everyday experiences—like feeling sound from a speaker or when talking while touching your throat.
- **Encourage Reflection:**
  - o Ask open-ended questions like, “Why do you think you felt that?” or “What would happen if there were no air or water?”

Builder

Objective:

Students will explore how sound travels through solids by building a string phone. This hands-on activity helps students understand sound vibrations, communication methods, and the science of how sound moves from one place to another.

Additional resources link: https://www.youtube.com/watch?v=PmmGSF7TXBI

Materials Needed:

- 2 paper cups
- A long piece of string (1–2 meters)
- 2 paper clips or tape
- Something to poke holes (like a pencil)

Activity Instructions:

1. Poke a small hole at the bottom of each paper cup.
2. Thread one end of the string through the hole in the first cup and tie a paper clip or knot it to keep it in place.
3. Repeat on the other side with the second cup.
4. Two students should each hold a cup and walk apart until the string is tight.
5. One student talks into the cup while the other places their ear on their cup to listen.

Possible strategies to implement:

- **Pair and Practice:** Assign students in pairs and rotate so every student has a turn speaking and listening.
- **Guided Observation:** Ask guiding questions like:
  - o "What do you feel if you touch the string?"
  - o "Does the string have to be tight or loose for the sound to travel?"
- **UDL Tip:** Allow students to sketch or act out what's happening inside the string. Use visuals to show sound waves traveling.
- **Scaffolded Support:** Give sentence starters: "I noticed that when I talked, the string...", or provide labels for parts of the model (cup, string, sound waves).
- **Class Discussion:** Compare experiences and reinforce the idea that sound moves by vibration and needs a medium to travel.

Final Prototype:

Final prototype video:

https://drive.google.com/drive/folders/1LZFl0M84ebjUlqqKPpyOk4fENppoSBbH

Rubric – My Smart LED Home!

Criteria	Needs Improvement (1 pt)	Developing (2 pts)	Proficient (3 pts)	Excellent (4 pts)
Model Construction	Model is incomplete or not working	Cups are connected but string is too loose or tangled	String phone works with some help from teacher	String phone is fully functional and well built independently
Understanding of Sound Travel	Does not show understanding of how sound moves	Partial understanding shown through explanation or action	Understands that sound moves through string	Clearly explains that sound travels as vibrations through solids
Teamwork & Communication	Did not cooperate or share tasks	Sometimes cooperated but needed reminders	Worked well with partner, shared tasks fairly	Actively supported partner and worked together the whole time
Observation & Reflection	Gave no observations or ideas	Gave a basic answer with teacher prompting	Shared a clear observation of how the phone worked	Gave thoughtful observations and made connections to sound/vibration
Effort & Participation	Needed constant redirection	Participated with occasional reminders	Stayed focused and followed instructions	Fully engaged, asked questions, and showed curiosity throughout

Now I Can

Objective:

Students will demonstrate their understanding of how sound is made through vibration, how it can be felt and observed, and how voice commands can interact with technology like LEDs. This reflection helps reinforce key science and technology concepts introduced through hands-on activities.

Activity Instructions:

1. Display the “Now I Can” checklist on the board or distribute printed versions.
2. Read each statement aloud one at a time and ask students to put a checkmark if they feel confident about it.
3. Pause after each sentence and allow students to share an example or explain what they remember (e.g., “What did we use to see sound move?”).

4. Encourage peer sharing — students can explain to a partner how they completed a specific activity.
5. Celebrate learning progress and clarify any concepts that students are unsure about by revisiting related parts of the lesson.

Relation to Standard

1. Next Generation Science Standards (NGSS):

1-PS4-1:
Plan and conduct investigations to provide evidence that vibrating materials can make sound and that sound can make materials vibrate.

- Students observe how sound travels and makes objects move (e.g., sprinkles jumping).

3-PS2-2 (extension):
Make observations and/or measurements of an object's motion to provide evidence that a pattern can be used to predict future motion.

- The string phone shows how sound waves can travel predictably through a medium.

2. ISTE Standards for Students:

Empowered Learner (1a, 1c):

- Students set learning goals by checking "Now I Can" progress and using tools like voice-controlled LEDs.

Innovative Designer (4a, 4d):

- Students use design thinking to create a string phone and explore sound transmission.

3. CSTA Computer Science Standards (Grades K–2):

1A-CS-02:
Use appropriate terminology in identifying and describing the function of common physical computing components (e.g., LEDs, sensors).

- The LED activity reinforces understanding of inputs and outputs using voice recognition.

4. UN Sustainable Development Goals (SDG) Alignment:

SDG 4 – Quality Education:

- Promotes inclusive, hands-on, inquiry-based learning experiences.

SDG 9 – Industry, Innovation, and Infrastructure:

- Introduces smart tech and communication methods (voice commands, sound travel).

SDG 10 – Reduced Inequalities:

- Supports accessibility concepts (voice interfaces for assistive tech).

Mission Innovate

Design Another Smart Room

Overview:

This open-ended project challenges students to extend their understanding of inclusive smart design by creating a new tool for a different room in the home. Students will work as young engineers to make a space easier for someone who is blind, deaf, or has another disability. This encourages empathy, design thinking, and hands-on prototyping.

Project Duration: 1–2 sessions

Teacher's Role:

- Facilitator of Inquiry:
- Begin by reviewing the mission prompt with the class. Ask students to choose a room (not the same as in the first mission) and think about what's tricky for someone with a disability. Guide them to brainstorm problems and voice-activated or alternative solutions.
- Encourager of Empathy and Creativity:
- Lead a discussion using the critical thinking questions. Help students imagine how daily life is affected by blindness or hearing loss. Inspire them to think of kind, smart, and helpful features to support independence.
- Supporter of Student Work:
- Let students design and build independently or in pairs. Offer feedback only when needed, reminding them to apply what they've learned about voice control, recycled materials, and safety.
- Guide for Reflection and Communication:
- Before presenting, support students in explaining how their model helps people with disabilities. Use simple sentence starters if needed ("My design helps because..."). Invite respectful listening during presentations.
- Assessment Guide:
- Use a simple rubric (focused on design idea, usefulness, creativity, and clarity) to assess each team's final prototype.

Implementation Tips:

- Encourage students to build different rooms than the one they used in their first project.
- Offer recycled or classroom-safe materials like cardboard, straws, string, LED or buzzer kits, markers, etc.
- Help students brainstorm features like vibrating alerts, braille stickers, or sound-activated fans or switches.
- Support teamwork while allowing students to make individual design choices.
- Celebrate inclusive thinking and all ideas that focus on helping others.